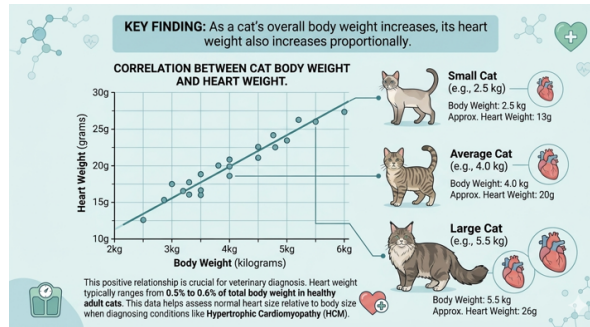


17 Linear regression and correlation



Gemini figure (use with caution - notice where the small cat is positioned)

Consider a simple example in which you have measurements of body weight and heart weight for a sample of cats.

Example: cat's body weight and heart weight

Assume we are interested to know whether a cat's body weight and heart weight are associated, specifically, whether bigger cats tend to have bigger hearts.

```
library(ggplot2)
library(MASS)
library(patchwork)

# Plot 1: Simple scatter plot
p1 <- ggplot(cats, aes(x = Bwt, y = Hwt)) +
  geom_point(color = "red", alpha = 0.6) +
  labs(x = "Body weight (kg)",
       y = "Heart weight (g)",
       title = "Cat's body weight vs. Heart weight") +
  theme_minimal()
```

p1



The scatterplot suggests a general linear trend. In fact, we see a **positive association** between body weight and heart weight, indicating that the heavier the body, the heavier the heart. Then it is natural to try to describe the strength of the linear association. One way to measure such strength of linear association is with the **correlation coefficient** defined as follows:

$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

The correlation coefficient is unit free and always between -1 and 1. The sign of the correlation indicates the sign of the relationship. In our example

```
sc<-cor(cats$Bwt,cats$Hwt)
```

sc

```
[1] 0.8041274
```

which confirms the positive **linear** association.

Things to remember:

- Our correlation is based on a sample, therefore we obtain a sample correlation and it is an estimate of the population correlation (often denoted by ρ (Greek letter rho)).
population parameter r estimator of ρ
- Any observed association between two variables does not imply causal connection between them.

17.1 Testing correlation

We can test $H_0 : X$ and Y are uncorrelated in the population, with a t-test with

test statistic

$$t_s = r \sqrt{\frac{n-2}{1-r^2}} \sim t_{n-2}$$

with $n - 2$ degrees of freedom

In our example:

```
n<-nrow(cats)
n
```

```
[1] 144
```

```
ttest<-sc*sqrt((n-2)/(1-sc^2))
pval<-2*(1-pt(ttest,n-2))
pval
```

```
[1] 0
```

Why $df = n - 2$? Any two points determine a straight line, however with 2 points we do not have information about the inherent variability in the scatter of the points.

```
cor.test(cats$Bwt,cats$Hwt)
```

Pearson's product-moment correlation

*

```
data: cats$Bwt and cats$Hwt
t = 16.119, df = 142, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.7375682 0.8552122
sample estimates:
      cor
0.8041274
```

A confidence interval can be obtained, however it is a little tricky. The distribution of r is sort of skewed. So we make use of a quantity known as Fisher's transformation of r :

$$z_r = \frac{1}{2} \ln \left[\frac{1-r}{1+r} \right]$$

pop. quantity

$$Z = \frac{1}{2} \ln \left[\frac{1-\rho}{1+\rho} \right]$$

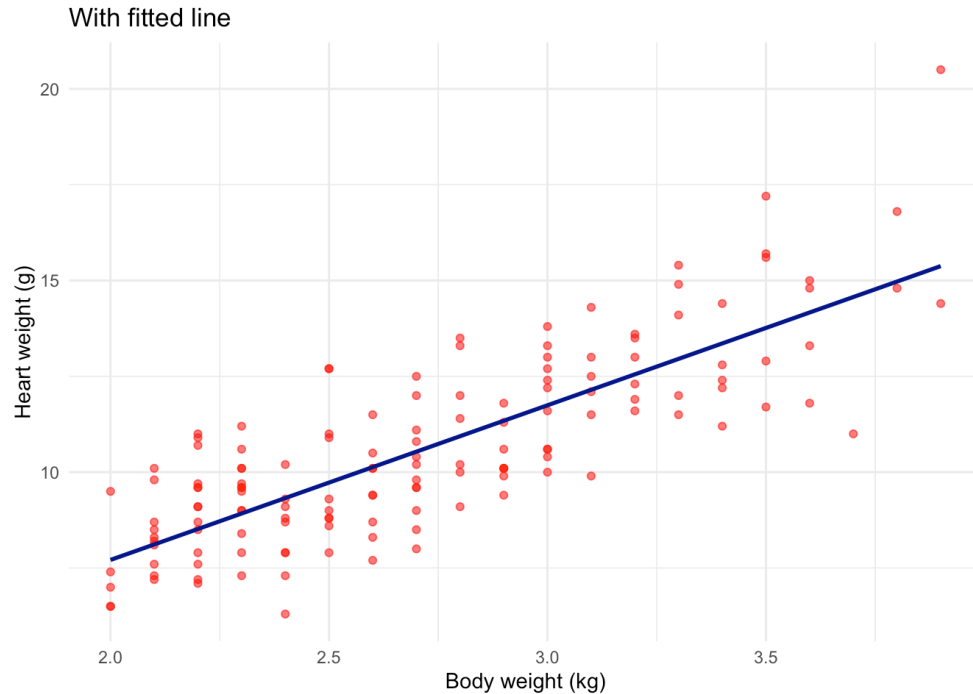
z_r behaves more like normally distributed. A 95% confidence interval can be constructed as follows:

$$z_r \pm 1.96 \frac{1}{\sqrt{n-3}}$$

But how do we form a confidence interval for ρ ?

$$\frac{1}{2} \ln \left[\frac{1 + \rho}{1 - \rho} \right] = z_r \pm 1.96 \frac{1}{\sqrt{n - 3}}$$

Given that in our first example we observe a clear linear relationship between body weight and heart weight, we can superimpose a line on the plot. This is called the **least-squares line** or **fitted regression line** of Y on X .



We will now learn how to find and interpret the line that best summarizes their linear relationship.

17.2 Linear Regression

Linear regression is a machine learning and statistical method that has been widely used for a long time to predict quantitative continuous responses. Although relatively simple, it is a very important model for understanding more complex machine learning models, which can be seen as generalizations of it.

Formally, suppose we want to understand how one or more variables — called predictors — relate to an outcome of interest — called the response variable. In our example, body weight (X) is the predictor and heart weight (Y) is the response. A few key questions naturally emerge:

- Is there a relationship between body weight and heart weight?
- How strong is that relationship?
- How large is the effect — for each additional kilogram of body weight, how many grams does the heart weight increase?
- How accurately can we predict heart weight from body weight?
- Is the relationship linear, or does it follow a more complex pattern?

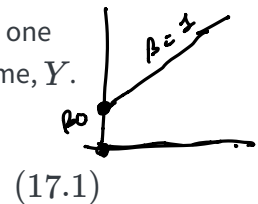
- When multiple predictors are available — such as body weight, sex, and age — which ones are truly associated with heart weight, and is there any interaction among them?

Linear regression allows us to address all of these questions in a rigorous and interpretable way. And while our example is biological, the same framework applies broadly — from medicine and ecology to economics and engineering.

17.3 Simple Linear Regression

Simple linear regression refers to the most basic model when there is only one predictor, X , and how that predictor is linearly related to a specific outcome, Y . Mathematically, this linear relationship is **modeled** as follows:

$$\text{data} \rightarrow Y \approx \underbrace{\beta_0}_{\text{intercept}} + \underbrace{\beta_1 X}_{\text{slope}} \leftarrow \text{data}$$



[Equation 17.1](#) is interpreted as we are regressing Y on X , and β_0 , and β_1 are model *coefficients* or *parameters* of the model.

- β_0 represent the *intercept*
- β_1 the *slope* in the linear model.

The main objective of linear regression is to use training data (in the previous example, X represents measures of body weight for the sample of cats, and Y observations of heart weight), to produce estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ for the model coefficients that better approximate the relationship between the predictor and outcome. That is:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X. \quad (17.2)$$

[Equation 17.2](#) $\hat{y}$ indicates a prediction of Y (body weight) on the basis of X (=x, heart weight), and the symbol $\hat{}$ denotes a predicted value.

Note

Example: Cat's Heart Weight

Linear regression of cat's heart weight on body weight, using the dataset *cats* from the MASS library in R.

$$\hat{\beta}_0 = 0$$

$$\hat{y} = \hat{\beta}_1 X$$

```
model1 <- lm(Hwt ~ Bwt, data = cats)
```

$$\text{lm}(\text{Hwt} \sim \text{Bwt}, \text{data} = \text{cats})$$

Call:

```
lm(formula = Hwt ~ Bwt, data = cats)
```

Coefficients:

(Intercept)	Bwt
-0.3567	4.0341

Notice that the intercept is equal to -0.3567 , which we can ignore as it is negative and has no biological meaning (heart weight when body weight = 0). The slope is important in this model, because indicates that for every 1 kg increase in body weight, heart weight increases by about 4 grams.

In summary, body weight is a strong positive predictor of heart weight in cats. Larger cats tend to have proportionally heavier hearts.

```
summary(model1)
```

Call:

```
lm(formula = Hwt ~ Bwt, data = cats)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.5694	-0.9634	-0.0921	1.0426	5.1238

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3567	0.6923	-0.515	0.607
Bwt	4.0341	0.2503	16.119	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.452 on 142 degrees of freedom

Multiple R-squared: 0.6466, Adjusted R-squared: 0.6441

F-statistic: 259.8 on 1 and 142 DF, p-value: < 2.2e-16

Goodness of fit:

R-squared = 0.64 - indicates that body weight explains 64% of the total variation in heart weight.

p-value < 2.2e-16 - indicates that the relationship is statistically significant.

Residual standard Error: 1.452g - indicates how far off predictions typically are (we are typically wrong by about 1.42 grams)

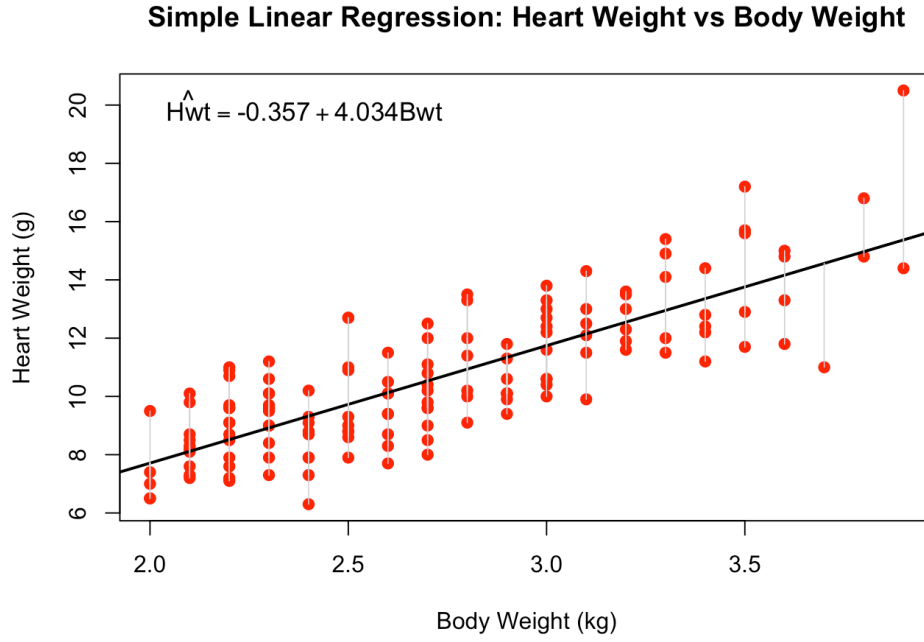
17.3.1 Estimating the coefficients

The coefficients β_0 and β_1 in the linear model are unknown. The goal is therefore to obtain estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ such that the linear model [Equation 17.1](#) fits the $n = 144$ data points as closely as possible.

There are several ways to measure *closeness*, however the most common approach involves minimizing the *least squares* criterion.

In least squares, we calculate the difference between the observed value y_i and the predicted value $\hat{y}_i$ obtained from the fitted model (from [Equation 17.2](#) above: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$). That is, $e_i = y_i - \hat{y}_i$ represents the *i-th residual* for observation *i*. We illustrate these residuals for the *cats* sample in the plot below,

shown as vertical lines from each observed data point (in red) to the fitted line.



Then, we define the *residual sum of squares* (RSS) as:

$$RSS = e_1^2 + e_2^2 + \cdots + e_n^2 \quad (17.3)$$

The *least squares coefficient estimates* $\hat{\beta}_0$ and $\hat{\beta}_1$ for simple linear regression are obtained by minimizing the RSS, which after some calculus yields:

$$\begin{aligned} \hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = r \left(\frac{s_y}{s_x} \right), \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}, \end{aligned} \quad (17.4)$$

where $\bar{y} \equiv \frac{1}{n} \sum_{i=1}^n y_i$ and $\bar{x} \equiv \frac{1}{n} \sum_{i=1}^n x_i$ are the sample means.

17.3.2 Assessing the accuracy of the coefficient estimates

A natural question about the linear regression model is how accurate the coefficient estimates are — that is, how far $\hat{\beta}_0$ and $\hat{\beta}_1$ are from the true values β_0 and β_1 . To answer this question, we compute the *standard errors* associated with $\hat{\beta}_0$ and $\hat{\beta}_1$ using the following formulas:

$$SE(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right], \quad SE(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (17.5)$$

where $\sigma^2 = \text{Var}(\epsilon)$ is the variance of the error terms, which is unknown but can be estimated empirically via the RSE defined as follows:

$$RSE = \sqrt{\frac{RSS}{n - 2}}$$

Strictly, these equations are valid only when the errors ϵ_i for each observation have constant variance σ^2 and are uncorrelated, but in practice they provide a good approximation.

Notice that unlike e_i , which are the observed residuals obtained from the fitted model, ϵ_i are the underlying population errors that arise when approximating the relationship between X and Y through a linear model, and are therefore unobserved.

Once we obtain the standard errors of the estimates, we can construct *confidence intervals*. A 95% confidence interval is defined as a range of values such that, with 95% probability, the range will contain the true unknown value of the parameter. Confidence intervals are constructed from the data and have the property that if we take repeated samples and construct a confidence interval for each sample, 95% of those intervals will contain the true unknown value of the parameter. For the linear regression model, the 95% confidence intervals for β_0 and β_1 are approximated as:

$$\left[\hat{\beta}_0 - 2 \cdot SE(\hat{\beta}_0), \hat{\beta}_0 + 2 \cdot SE(\hat{\beta}_0) \right] \quad (17.6)$$

$$\left[\hat{\beta}_1 - 2 \cdot SE(\hat{\beta}_1), \hat{\beta}_1 + 2 \cdot SE(\hat{\beta}_1) \right] \quad (17.7)$$

In the previous example, the standard errors for the intercept and slope were 0.6923 and 0.2503, respectively. The associated confidence intervals are therefore:

```
confint(model1)
```

```

                2.5 %    97.5 %
(Intercept) -1.725163  1.011838
Bwt          3.539343  4.528782
```

We are thus 95% confident that for every 1 kg increase in body weight, the true average increase in heart weight is between 3.54 and 4.53 grams.

Standard errors are also useful for testing the most common hypothesis in regression — that there is no relationship between X and Y (i.e., that a cat's body weight is not associated with its heart weight). Formally, the *null hypothesis* states that there is no relationship between X and Y , versus the *alternative hypothesis* that there is some relationship between X and Y . Mathematically, this corresponds to testing:

$$\begin{aligned} H_0 : \beta_1 &= 0 \\ \text{versus} \\ H_a : \beta_1 &\neq 0 \end{aligned} \quad (17.8)$$

To test the null hypothesis, we need to determine how far $\hat{\beta}_1$ is from zero to be

confident that β_1 is nonzero. This depends on the accuracy of the estimate, given by $SE(\hat{\beta}_1)$. The smaller $SE(\hat{\beta}_1)$ is, the more likely that β_1 differs from zero; conversely, the larger the standard error, the larger $\hat{\beta}_1$ needs to be in order to reject the null hypothesis. In practice, we rely on a *t-statistic*, given by

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}, \quad (17.9)$$

which measures how many standard deviations $\hat{\beta}_1$ is from zero. If there is no relationship between X and Y , we expect t to follow a *t*-distribution with $n - 2$ degrees of freedom. The larger the observed value of t relative to the theoretical *t*-distribution, the lower the probability that $\beta_1 = 0$. This probability is called the *p-value*, which in our example was practically zero ($< 2 \times 10^{-16}$), providing strong evidence that $\hat{\beta}_1 \neq 0$, or equivalently that it is *statistically significant*. A small *p-value* indicates that it is unlikely to observe such a substantial association between the predictor and the response by chance alone. We therefore reject the null hypothesis when the *p-value* is sufficiently small — typically below 0.05 or 0.01.